Rabeya=43

ch=4

Polymerization

- Step Polymerization
- Addition Polymerization
  - Free Radical Polymerization
  - Ionic Polymerization
    - A. Cationic Polymerization
    - B. Anionic Polymerization

Q Define Step growth (condensation) polymerization with suitable example (1.5)

Step Growth (Condensation) polymerization

Tu-18

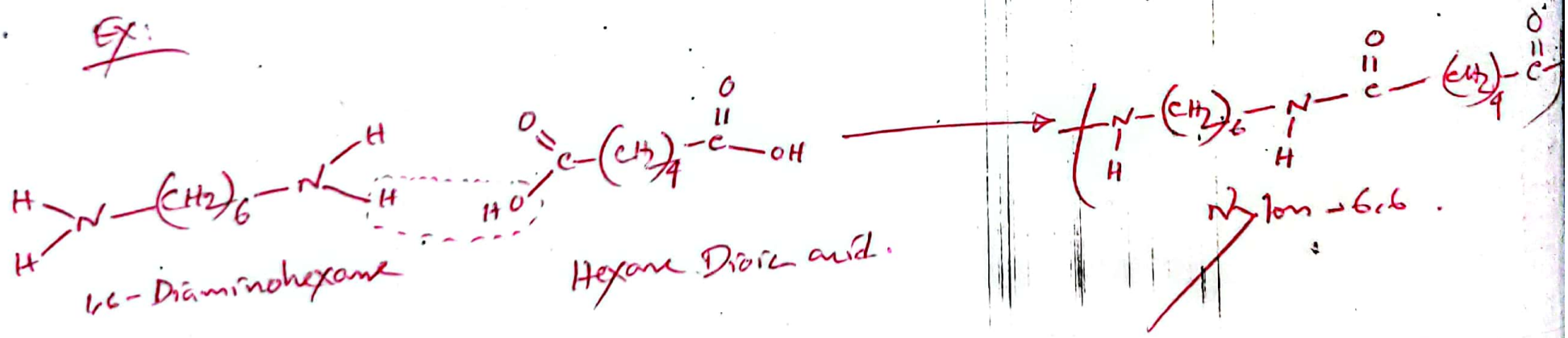
A condensation takes place between two bilyfunctional molecules to produce one larger polyfunctional molecule with the *possible* elimination of a small molecule such as H_2O, NH_3, HCl, etc. The reaction continues until one of the reagents is used up.

Common types:

- Polyamides
- Polyesters
- Polycarbonates
- Polyurethanes

example (Tu-20)

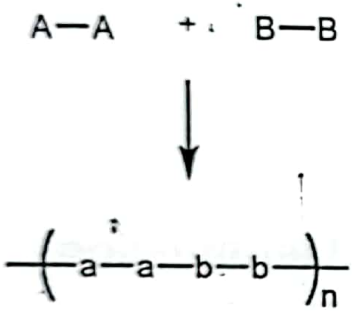
Ex:



Step Growth Polymerizations

Kabir
Che-46
AB

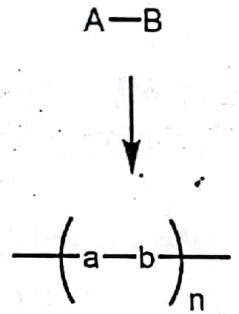
Bifunctional monomers



A-A + B-B ⟶ A-AB-B ⟶ B-BA-AB-BA-AB-BA-AB-B

1-mer 1-mer 2-mer poly-mer

AB monomers (Self condensation)



A-B A-BA-B $[A\text{-}B]_n$

1-mer 2-mer poly-mer

1. Difunctional monomers ⟹ Linear polymers

   AB type, AA and BB

2. Polyfunctional monomers ⟹ Network polymers

   Functionality > 2

3. Polymers retain their functionality as end groups at the completion of polymerization

4. Single reaction (cf. initiation, propagation, and termination in chain-reaction polymerization)

5. M.W. increases slowly

   Carothers equation

$$\overline{DP} = \frac{1}{1-p}$$

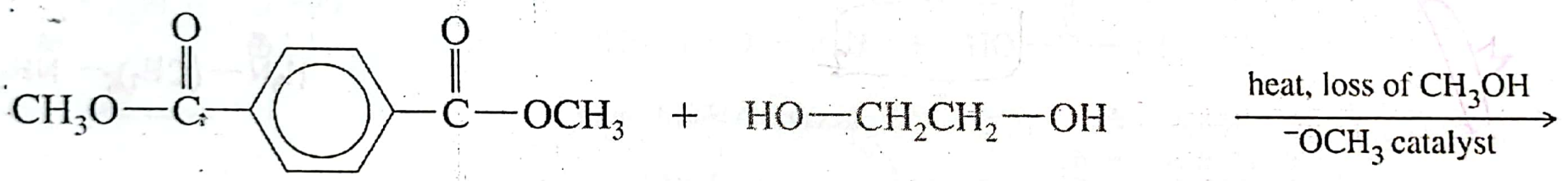
p = extent of reaction

6. High-yield reactions and an exact stoichiometric balance are necessary to obtain high-m-w linear polymer

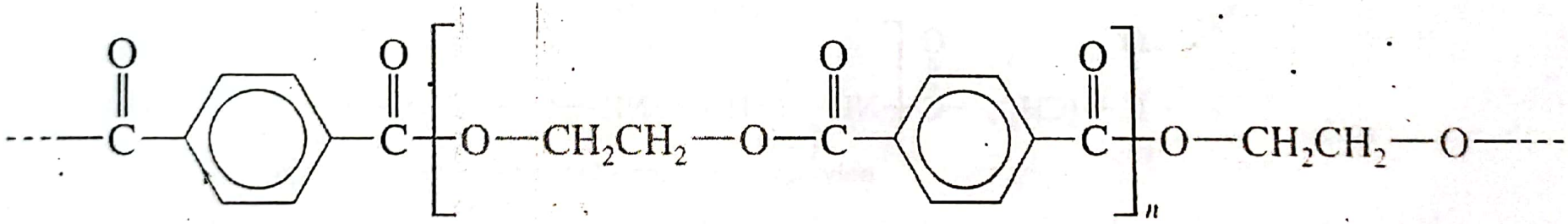
Kabir che-46

Polyesters (preparation by name প্রস্তুত 2টা)

- Polymer of terephthalic acid and ethylene glycol.
- Made by the transesterification of the methyl ester.



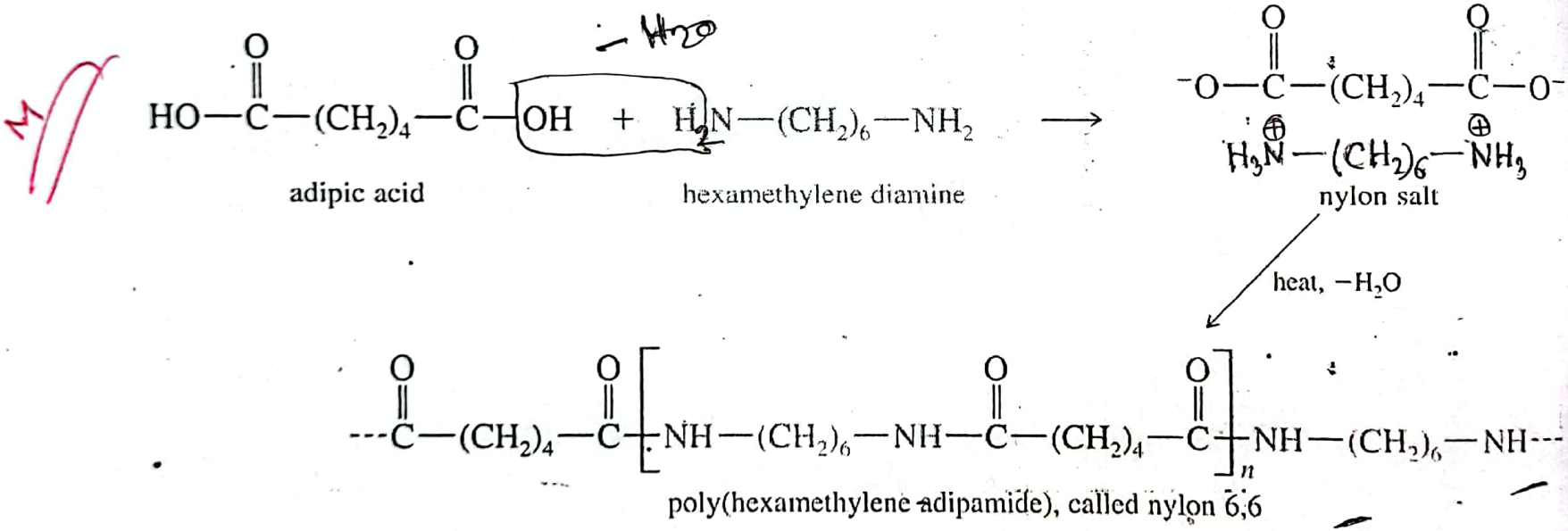
dimethyl terephthalate ethylene glycol



poly(ethylene terephthalate) or PET, also called Dacron® polyester or Mylar® film

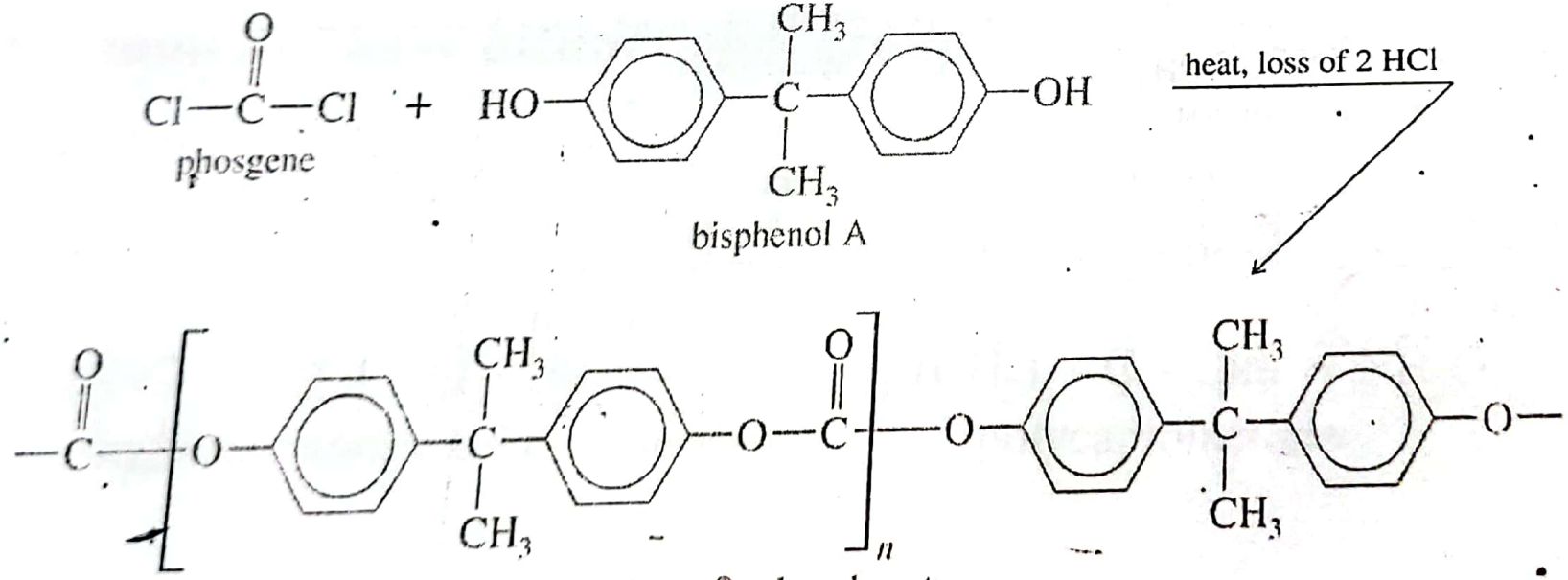
Polyamides: Nylon

Usually made from reaction of diacids with diamines, but may also be made from a single monomer with an amino group at one end and acid group at other.



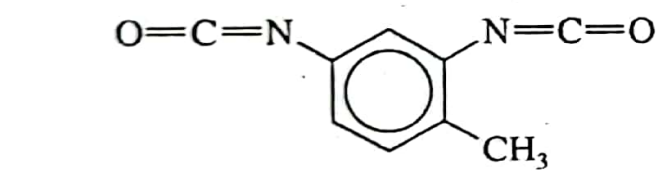
Polycarbonates

- Esters of carbonic acid.
- React phosgene with bisphenol A to obtain Lexan for bulletproof windows.



Polyurethanes

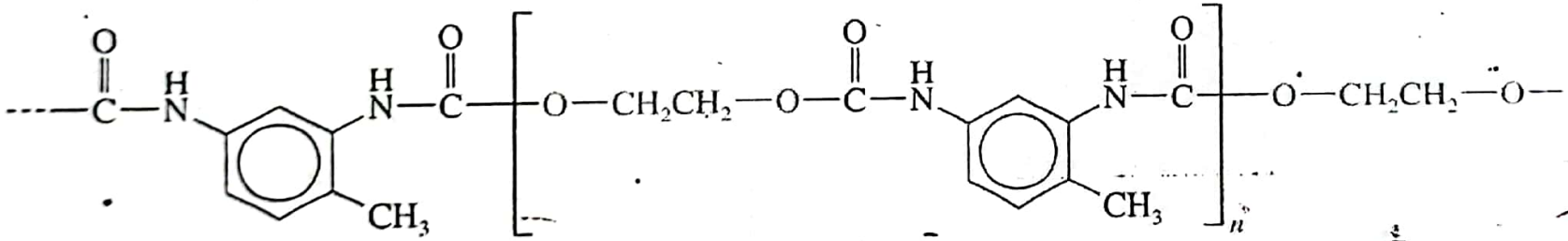
- Esters of carbamic acid, R-NH-COOH.
- Urethanes are prepared by reacting an alcohol with isocyanate.
- Polyurethanes are prepared by reacting a diol with a diisocyanate.



toluene diisocyanate

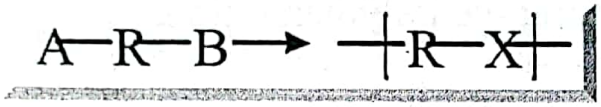
+ HO—CH_2CH_2—OH

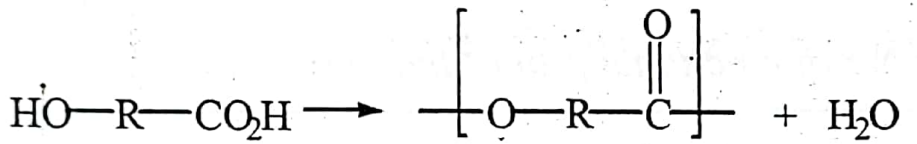
ethylene glycol

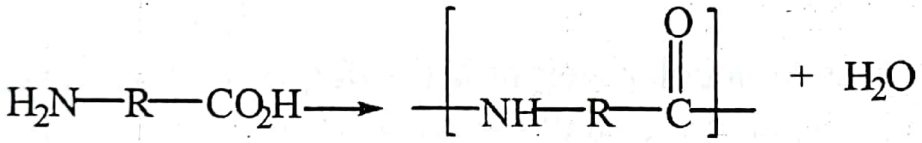


a polyurethane

One having both reactive functional groups in one molecule

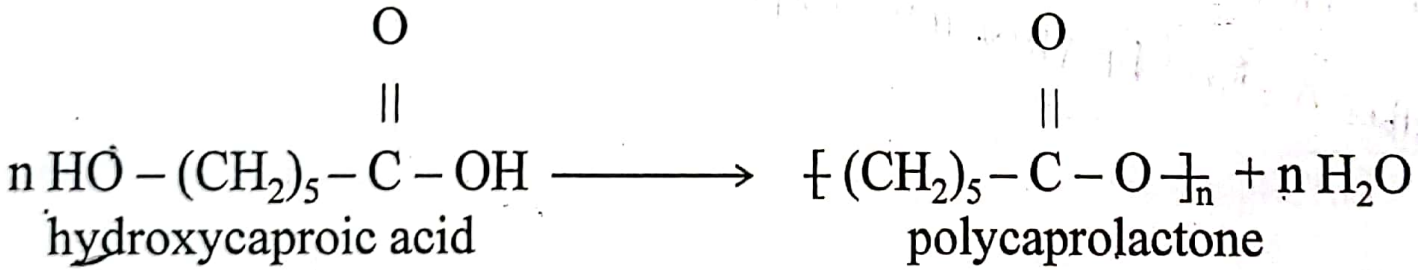






Monomer to have difunctional group

Self condensation



MW in Step-Growth polymerization

Polymer MW in step-growth polymerization is determined by the fractional conversion p, of the monomer during the polymerization.

Extent of reaction *p*:

$$p = \frac{\textit{number of functional groups that have reacted}}{\textit{number of functional groups originally present}}$$

One way to express molecular weight is the *degree of polymerization*- number of repeating units in the chain. Usually *average degree of polymerization* ($\overline{X_n}$ *or* $\overline{DP}$) is u as polymerization mechanism yields distribution of molecular weight.

MW কিভাবে express করা যায়?
Carother ও Flory DP দিয়ে MW express করা যায়।

Carothers equation 2016 * *

Consider a flask of monomer ....

If there are N_o molecules in the flask at time = 0 and N remaining at time t,

then the $\overline{DP}$ at time t is the average degree of polymerization... must just be, N_0/N.

The amount reacted is N_0-N and we can define p as the "conversion" or fraction reacted then as

$$p = (N_o - N) / N_o \text{ or}$$

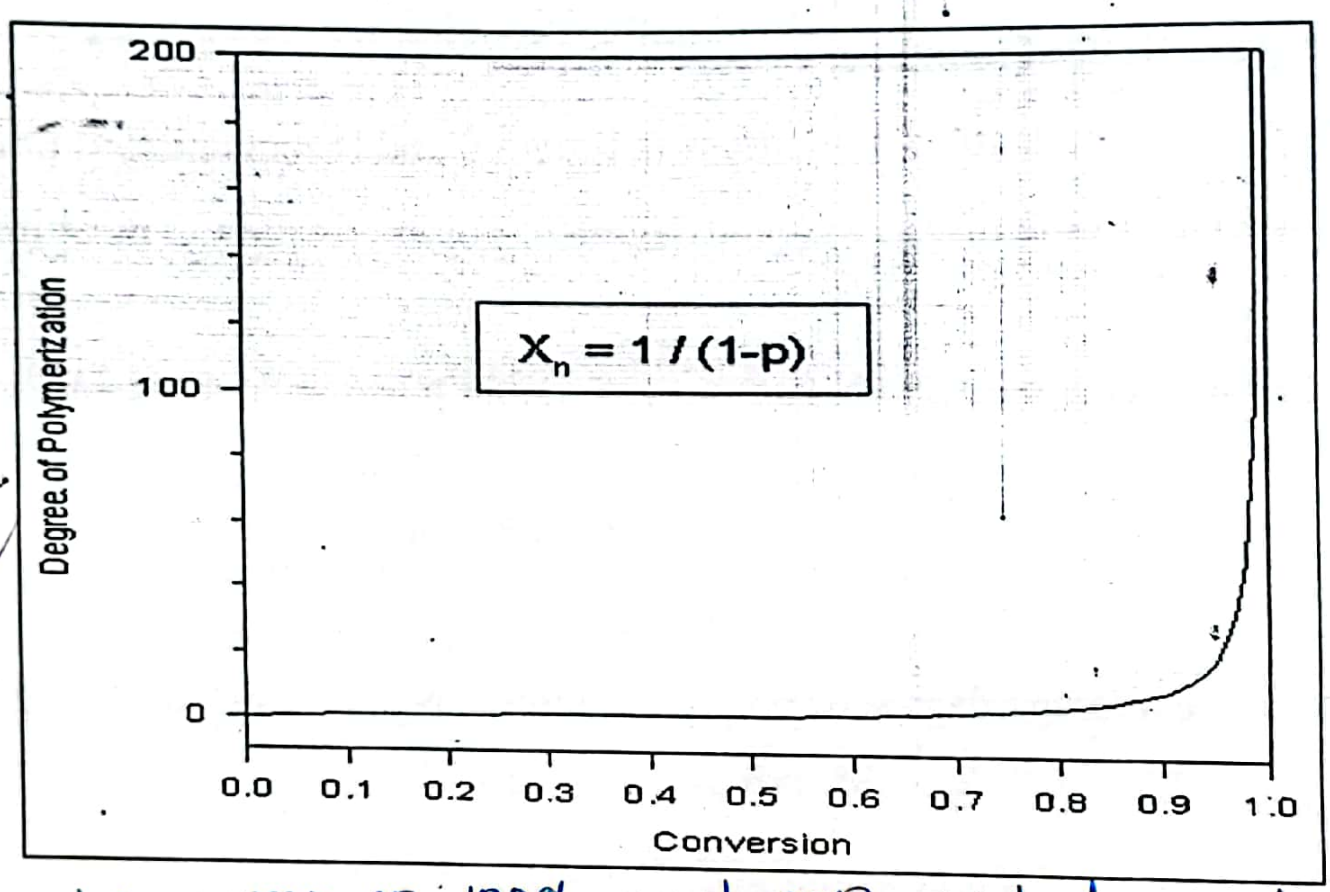
$$N = N_o(1 - P)$$

If $\overline{DP}$ is the average degree of polymerization... N_0/N ....substituting gives

$$\overline{DP} = 1/(1-P)$$

and for P = 0.98 (98% conversion), $\overline{DP}$ = only 50!

Carothers equation - relates/calculate molecular weight to the extent of a step growth polymerisation



Meaning of the graph: monomer 100% react করলে high degree of polymer পাওয়া যায়। initially ⇒ Dp কম। end stage ⇒ বেশি।

Number-average degree of polymerisation, $\overline{X}_n = N_0/N$

Conversion, $p = (N_0-N)/N_0 \longrightarrow N = N_0(1-p)$

Carothers equation

$$\overline{X}_n = 1/(1-p)$$

Number-average molecular weight: $\overline{M}_n = \overline{X}_n \times M_0 = M_0/(1-p)$
where M_0 is the molecular weight of the repeating unit

The weight-average degree of polymerisation, $\overline{X}_w = (1+p)/(1-p)$

Step – growth polymerization : MW distribution

Molecular weight distribution (polydispersity index) is $= \dfrac{\overline{X}_w}{\overline{X}_n} = \dfrac{(1+p)/(1-p)}{1/(1-p)} = (1+p)$

As $p \rightarrow 1$, MWD =2 for any linear step – growth polymerization .

MW-এর polydespersity মান বা ভিন্নতা ০1 MW মান. অসাম্য।

KINETICS OF STEP GROWTH POLYMERIZATION

WHY BOTHER ?

- How long does it take to make polymer ?
- Can we speed up the reaction ?
- What is the relationship between kinetics and the Mol. Wt. Of the product ?

8. Discuss the kinetics of step-reaction polymerization in the presence and absence of acid catalyst.

March 2016

KINETICS OF POLYCONDENSATION 2016

Let us consider the reaction below

$$\mathbf{A-A + B-B \longrightarrow A-AB-B}$$

Consider a polyesterification of bifunctionalmonomers in which initially equal amounts of di-basicacid and glycol are present.

$$HOOC-R_1-COOH + OH-R_2-OH \longrightarrow \text{-----} \overset{\overset{O}{\|}}{CO} \text{-----}$$

Case 1: Acid Catalyzed Polyesterification :

If the polyesterification is first order with respect to each functional group reactant, A and B, then the rate expression is

$$\text{Reaction Rate} = -\frac{d[A]}{dt} = k[A][B] \quad \text{--------(1)}$$

Given the fact that an exact stoichiometric balance is usually a necessary condition for achieving high molecular weight, we can equate [A] and [B] and the rate expression becomes,

$$-\frac{d[A]}{dt} = k[A]^2 \quad \text{---------- (2)}$$

or, by integration,

$$\frac{1}{[A]} - \frac{1}{[A_0]} = kt \quad \text{----------(3)}$$

At any particular , t, $\overline{DP}$ is equal to the ratio of monomer molecules present initially to the total number at that time; that is

$$\overline{DP} = \frac{[A_0]}{[A]} \quad \text{---------- (4)}$$

Now the Carothers equation is

$$\overline{DP} = \frac{1}{1-p} \quad \text{---------- (5)}$$

Now combining eq. (4) and (5) and solving for [A],

$$[A] = [A_0](1- p) \text{ ---------- (6)}$$

By substituting this value into eq. (3), we obtain

$$\frac{1}{[A_0](1- p)} - \frac{1}{[A_0]} = kt \text{ ---------- (7)}$$

Since $\overline{DP} = 1/(1- p)$

$$\frac{\overline{DP}}{[A_0]} - \frac{1}{[A_0]} = kt \text{ ---------- (8)}$$

Or

$$\overline{DP} = [A_0]\, kt + 1$$

Case 2: No Acid Catalyst Added :

In the absence of added acid, the carboxylic acid assumes the role of catalyst and the rate expression becomes,

$$-\frac{d[A]}{dt} = k[A]\,[A][B]$$

That is

$$-\frac{d[A]}{dt} = k[A]^2[B] \quad \text{----------} \quad (1)$$

Again assuming [A] and [B] are equal,

$$-\frac{d[A]}{dt} = k[A]^3 \quad \text{----------} \quad (2)$$

Integration then gives

$$\frac{1}{[A]^2} - \frac{1}{[A_0]^2} = 2kt \quad \text{----------} \quad (3)$$

Substituting for [A] as above and rearranging

since, $[A] = [A]_0 (1-p)$

$$\frac{1}{[A_0]^2 (1-p)^2} - \frac{1}{[A_0]^2} = 2kt \quad \text{----------} \quad (4)$$

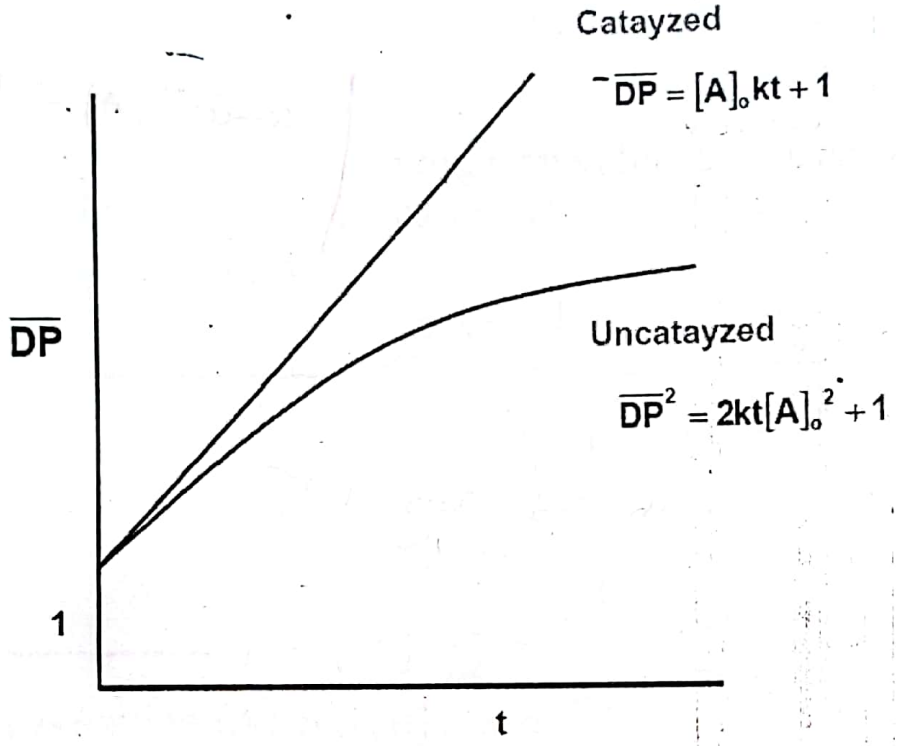
Or

Multiplying both side by $[A]_0^2$

$$\frac{1}{(1-p)^2} = 2kt\,[A_0]^2 + 1 \quad \text{----------} \quad (5)$$

and

$$\overline{(DP)^2} = 2kt\,[A_0]^2 + 1$$



The following conditions must be fulfilled in step-growth polymerizations for higher molecular weight:

- high conversions
- high monomer purities
- high reaction yields
- stoichiometric equivalence of functional groups

* Exact stochiometric balance is necessary to achieve high M.W.

balance হলে → high polymer M.W
imbalance " → low M.W · limit করার condition.

Stoichiometric Imbalance

Despite the fact that an exact stoichiometric balance is necessary to achieve high molecular weights in most instants of step reaction polymerization, in some important application it is necessary to limit the molecular weight.

•Epoxy resins are **polyethers** that are processed to an **oligomer stage** and are subsequently converted to **network polymer** by appropriate reactions of terminal epoxyide groups.

•With **polyimides** for fiber applications, molecular weight must often be limited because **too high a viscosity** is detrimental to **extrusion of filaments through the fine holes of a spinneret.**

M
2016 (C)

Three ways to **limit M. W.** in step polymerization **

1. **To quench the polymerization reaction by rapid cooling when the desired M.W. is attained.**
2. **To use an excess of one monomer.**
3. **To use small amounts of monofunctional reactant**

Stoichiometric imbalance

When the nonstoichiometric amount of functional groups is used, the relation between $\overline{DP}$ and the reaction conversion can be modified by modification of the Carothers equation.

In order to do so let us introduce a factor, *r,* representing the *stoichiometric imbalance.*

If the number of A and B functional groups present at the beginning of AA + BB polymerization is N_A^0 and N_B^0 respectively,

$$r = \frac{N_A^0}{N_B^0}$$

By convention r is always less than unity.

p = reaction conversion, in this case fraction of A group that have reacted.

As reaction of each A group consumes one B group, fraction of B reacted at p is equal

$$pN_A^0 \text{ or } prN_B^0$$

The number of *unreacted* groups , N_A and N_B is given by

$$N_A = (1-p)\,N_A^0 \quad \text{and} \quad N_B = (1-pr)\,N_B^0 = (1-pr)\,\frac{N_A^0}{r}$$

At this point, the number of A and B end group is equal to $N_A + N_B$, and because there are two end groups per molecule, the number of molecular chains, N, is given by

$$N = \frac{1}{2}(N_A + N_B)$$

That is

$$N = \frac{1}{2}\left[(1-p)\,N_A^0 + (1-pr)\frac{N_A^0}{r}\right]$$

which reduces to

$$N = \frac{N_A^0}{2}\left(1 + \frac{1}{r} - 2p\right)$$

We know that one repeating unit is formed every time an A and B functional group react; therefore the total number of repeating units N_r is given by

$$N_r = \frac{1}{2}(N_A^0 + N_B^0)$$

Since, $r = \frac{N_A^0}{N_B^0}$

$$N_r = \frac{1}{2}\left(N_A^0 + \frac{N_A^0}{r}\right) = \frac{N_A^0}{2}\left(\frac{r+1}{r}\right)$$

The average degree of polymerization is equal to the number of monomeric units divided the number of chains; that is

$$\overline{DP} = \frac{N_r}{N} = \frac{(N_A^0/2)[(r+1)/r]}{N_A^0/2[1+(1/r)-2p]}$$

$$r = \frac{N_A^0}{N_B^0} < 0, \quad (N_A^0 < N_B^0)$$

Carothers equation reduces to

$$\overline{DP} = \frac{r+1}{r+1-2pr}$$

Now if $r = 1$

$$\overline{DP} = \frac{1}{1-p}$$

When monomer A is completely used up in the polymerization (that is $p = 1$) the equation becomes

$$\overline{DP} = \frac{1+r}{1-r}$$

If monofunctional reagent is added

Imbalance factor

$$r' = \frac{N_A^0}{N_B^0 + 2\,N^0_{B'}}$$

where $N^0_{B'}$ = number of monofunctional B groups

Factor 2 : each B' molecule is equally as effective as one excess BB monomer in limiting the M.W.

Molecular weight distribution

In any polymerization involving difunctional monomer, each step reaction links two monomer together; thus the number of monomer units in the polymer is always one more than the functional group.

If a polymer contains x monomer units, the number of A or B groups that have reacted is x-1.

The probability that x-1 A or B groups has reacted is p^{x-1} , where p is the reaction conversion, defined previously as

$$p = \frac{N_0 - N}{N_0}$$

The probability of finding an unreacted group is

$$1 - p$$

The probability of finding a molecule containing x units and an unreacted A or B group is then

$$p^{x-1}(1-p)$$

If N= total number of molecule present at time t, the fraction that contains x units N_x, is given by

$$N_x = N\,p^{x-1}(1-p)$$

As $N/N_0 = 1-p$ (Carother equation), we can write the expression for N_x as

$$N_x = N_0(1-p)^2\,p^{x-1}$$

Where N_0 = initial number of monomers and the relationship defines number or mole fraction distribution at t.

The weight fraction w_x, of molecules containing x units is equal to the ratio of the mass of those molecules divided by the total mass; that is

$$w_x = \frac{xN_xM_0}{N_0M_0} = \frac{xN_x}{N_0}$$

Where M_0 is the mass of the repeating unit.

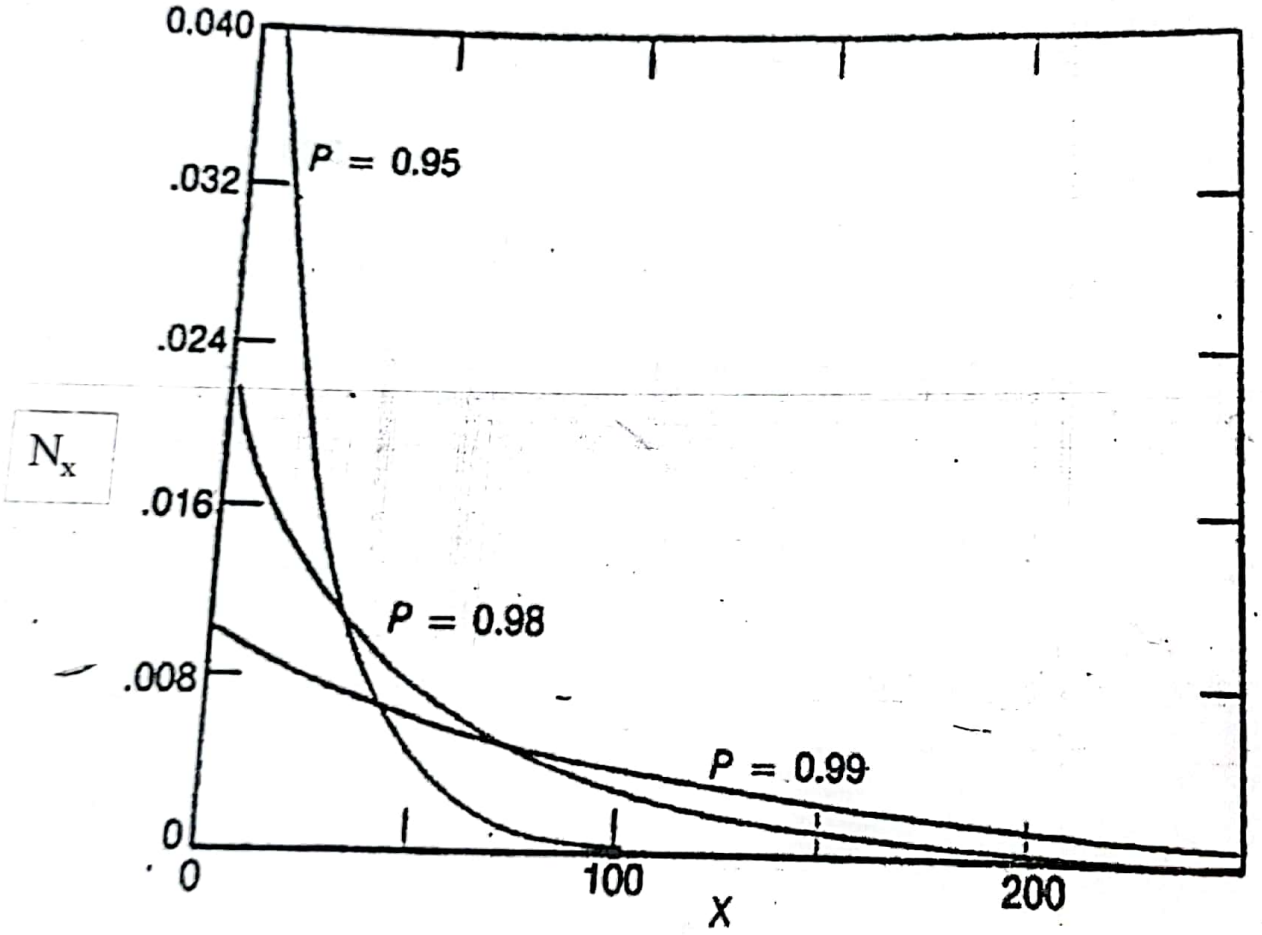
As, $N_x = N_0(1-p)^2 p^{x-1}$

We can write ,

$$w_x = x(1-p)^2 p^{x-1}$$

Which is the weight fraction distribution.

Mole fraction distribution in linear step-reaction polymerization



: Weight fraction distribution in linear step-reaction polymerization

